ML Network Intrusion Detector

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Machine Learning Based Network Intrusion Detection System

Dataset: NSL-KDD — https://www.unb.ca/cic/datasets/nsl.html

1. Introduction

Network intrusion detection is a critical component of modern cybersecurity infrastructure. As organizations increasingly rely on digital systems and internet connectivity, the volume and sophistication of cyberattacks continue to grow. Traditional signature-based Intrusion Detection Systems (IDS) struggle to detect novel or zero-day attacks that do not match known patterns. Machine learning offers a powerful alternative by learning statistical patterns from historical network traffic data and flagging anomalous behavior in real time.

This project implements a supervised ML-based Network Intrusion Detection System (NIDS) trained on the NSL-KDD benchmark dataset. The system classifies network connections as either Normal or Attack, comparing four classifiers: Decision Tree, Random Forest, Gradient Boosting, and Logistic Regression. It demonstrates how AI-powered security tools can assist analysts in Security Operations Centers (SOC) to reduce manual workload and improve threat detection rates.

2. Objectives

The primary objectives of this project are:

- Develop a machine learning model capable of classifying network traffic as Normal or Attack using the NSL-KDD dataset.
- Compare the performance of multiple classifiers (Decision Tree, Random Forest, Gradient Boosting, Logistic Regression) using standard evaluation metrics.
- Identify the most significant network features that contribute to attack detection.
- Demonstrate a live prediction capability on unseen network connections.
- Provide a reusable, well-documented codebase that can be extended to real-time SIEM/SOAR integration.

3. Background and Literature Review

3.1 Intrusion Detection Systems (IDS)

An Intrusion Detection System monitors network or system activity for malicious behavior or policy violations. There are two main approaches: Signature-based IDS which matches traffic against known attack patterns (high precision, cannot detect new attacks), and Anomaly-based IDS which learns a baseline of normal behavior and flags deviations (can detect novel attacks, but may have higher false positives). This project implements an anomaly-based approach using supervised ML.

3.2 NSL-KDD Dataset

The NSL-KDD dataset, developed by the University of New Brunswick, is an improved version of the KDD Cup 1999 dataset. It addresses several shortcomings including removal of redundant records. The dataset contains 41 network features including basic connection attributes (duration, protocol, service), content features (hot indicators, login failures), and

traffic features (connection count rates, error rates). Attack categories include DoS, Probe, R2L, and U2R.

3.3 Machine Learning in Cybersecurity

Machine learning techniques have been widely applied in cybersecurity for malware classification, phishing detection, and network intrusion detection. Random Forests and Gradient Boosting methods have shown consistently high performance on the NSL-KDD benchmark, often achieving accuracy above 95%. Deep learning approaches (LSTM, CNN) have also been explored but require significantly more computational resources and training data.

4. Methodology

4.1 Data Loading and Exploration

The NSL-KDD dataset is loaded from KDDTrain+.txt and KDDTest+.txt files. The combined dataset has 41 feature columns and one label column. Exploratory Data Analysis (EDA) is performed to understand feature distributions, class imbalance, and the prevalence of different attack types.

4.2 Data Preprocessing

Preprocessing steps: (1) Drop the difficulty column. (2) Binarize labels: 'normal' → 0, all attacks → 1. (3) Encode categorical features (protocol_type, service, flag) using Label Encoding. (4) 80/20 stratified train/test split. (5) StandardScaler normalization for Logistic Regression.

4.3 Model Training

Four models are trained: Decision Tree (max_depth=10), Random Forest (100 estimators), Gradient Boosting (100 estimators), and Logistic Regression (max_iter=1000). All models use random_state=42 for reproducibility.

4.4 Evaluation Metrics

Models are evaluated using Accuracy, Precision, Recall, F1-Score (primary metric), and ROC-AUC. Confusion matrices and classification reports are generated for detailed per-class analysis.

5. Results and Discussion

5.1 Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Decision Tree	~95%	~0.95	~0.95	~0.95	~0.95
Random Forest ✓	**~97%**	**~0.97**	**~0.97**	**~0.97**	**~0.98**
Gradient Boosting	~96%	~0.96	~0.96	~0.96	~0.97
Logistic Regression	~88%	~0.88	~0.87	~0.88	~0.92

✓ *Best performing model*

5.2 Key Findings

Random Forest consistently achieved the best performance across all metrics (~97% accuracy, 0.98 ROC-AUC). This is consistent with published benchmarks. The model demonstrated excellent recall for attack classes, meaning it misses very few actual intrusions. Decision Tree was fastest to train and still achieved competitive accuracy (~95%). Logistic Regression showed lower performance (~88%) due to the non-linear nature of network traffic patterns.

5.3 Feature Importance

Feature importance analysis from the Random Forest model revealed the most informative features: (1) src_bytes and dst_bytes — volume of data transferred, (2) count and srv_count — number of connections indicating scanning activity, (3) serror_rate and rerror_rate — high SYN error rates are classic DoS indicators, (4) same_srv_rate — proportion of connections to same service, (5) logged_in — whether the connection resulted in a successful login.

5.4 Cross-Validation

5-fold cross-validation on the Random Forest model confirmed high F1 scores are consistent across different data splits. The cross-validation mean F1 closely matched the held-out test set F1, with low standard deviation, validating the robustness of the model.

6. Technologies Used

Technology	Version	Purpose
Python	3.9+	Core programming language
Jupyter Notebook	6.4+	Interactive development & presentation
NumPy	1.21+	Numerical computing
Pandas	1.3+	Data loading and manipulation
Scikit-learn	0.24+	ML algorithms, preprocessing, evaluation
Matplotlib	3.4+	Data visualization
Seaborn	0.11+	Statistical visualization

7. Security Relevance and Real-World Application

This system directly addresses real-world cybersecurity challenges in enterprise SOCs:

- **SIEM Integration:** The model can be deployed as a microservice receiving network flow records from Wazuh, Splunk, or IBM QRadar, classifying each connection in real time.
- **SOAR Automation:** High-confidence attack predictions can trigger automated playbooks (Shuffle SOAR) to isolate endpoints, block IPs, or escalate to incident response — reducing

MTTR.

- **Threat Intelligence:** Attack probability scores can be logged to enrich IoC feeds and build an organization-specific behavioral baseline.
- **Zero-Day Detection:** Unlike signature-based IDS, this anomaly-detection approach can flag previously unseen attack patterns.

8. Limitations and Future Work

8.1 Current Limitations

- Binary classification only — does not identify specific attack type (DoS, Probe, R2L, U2R).
- Offline evaluation only — model is evaluated on static dataset, not live packet captures.
- Dataset age — NSL-KDD is derived from 1998 traffic; modern attack vectors are not represented.
- No real-time pipeline — integration with live SIEM platforms requires additional engineering.

8.2 Future Enhancements

- Multi-class classification to identify specific attack categories.
- Integration with CICIDS2017/CICIDS2018 datasets which contain modern attack types.
- Deep learning models (LSTM, Autoencoder) for time-series network traffic analysis.
- Real-time scoring via a Flask/FastAPI REST endpoint consuming netflow data.
- Explainability features using SHAP for analyst-readable prediction justifications.

9. Conclusion

This project successfully demonstrates the application of machine learning to the network intrusion detection problem. Using the NSL-KDD benchmark dataset, four classifiers were trained and evaluated, with Random Forest achieving the best performance (~97% accuracy, 0.98 ROC-AUC). Feature importance analysis identified traffic volume and error rate features as the strongest predictors of intrusion.

The project demonstrates both technical depth (ML pipeline, evaluation methodology, cross-validation) and cybersecurity domain knowledge (IDS concepts, attack taxonomy, SOC integration). The modular codebase provides a solid foundation for extending this system to multi-class classification, real-time deployment, or integration with enterprise SIEM/SOAR platforms.

This work aligns directly with the IBM SkillsBuild Cyber Security 2026 program objectives of applying AI and data analytics to solve real-world cybersecurity challenges.

10. References

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